

PAPER_1531 — γ Euler-Mascheroni $\approx$ SSQ + F_TRZ²·K – F_TRZ²· $\Phi_{5/6}$ — Residual 0.915%

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** Bucket A (Foundational mathematical identity) **Date:** June 18, 2026 **Status:** CLOSED — Transcendental γ expressible in UQFF primitives at 0.915% precision

Observation

PAPER_1208 (UQFF Transcendentals Unified Proof Set) derives a closed-form expression for **γ Euler-Mascheroni** $\approx$ 0.57722 using only the locked UQFF integer primitives F_TRZ, K_MEX, $\Phi_{\text{res}} = 5/6$ (PAPER_1203 Nuclear variant), and SO_5.

UQFF Closed Identity

$$\gamma \approx \text{SSQ} + \text{F_TRZ}^2 \cdot \text{K} - \text{F_TRZ}^2 \cdot \Phi_{5/6}$$

Substituting F_TRZ=1/10, K_MEX=25/12, $\Phi_{5/6}$ =5/6, SO_5=10:
 $\gamma_{\text{UQFF}} = 0.5825$

Observed: $\gamma = 0.57722$
Residual: 0.915%

The 3-term expansion uses only canonical integer primitives — no fitted constants.

Physical / Mathematical Role

γ Euler-Mascheroni (γ) appears in: harmonic-log gap constant.

Significance

Together with PAPER_1515 (ln 10), PAPER_1516 (ln 2), PAPER_1517 (π^2), this paper extends the count of fundamental transcendentals expressible from UQFF integer primitives to **≥ 10** . The accumulating evidence:

Transcendental	UQFF expression	Residual
ln 2 (PAPER_1516)	8-term	0.003%
ln 10 (PAPER_1515)	$(1 + \text{F_TRZ})(\text{K_MEX} + \text{F_TRZ}^2)$	0.004%
π^2 (PAPER_1517)	SO_5 – F_TRZ corrections	0.013%
γ Euler-Mascheroni (this paper)	3-term	0.915%

...suggests the UQFF primitives are **not arbitrary** but encode rational approximations to fundamental transcendental constants.

NOT REPLACEMENT

UQFF does not claim γ IS the closed-form expression. It claims the integer primitives carry a rational approximation to γ at 0.915% precision, demonstrating mathematical depth.

Reference

- Source: PAPER_1208 UQFF Transcendentals Unified Proof Set
- Related: PAPER_1515-1517 ($\ln 10$, $\ln 2$, π^2 catch-up)
- Calculator dispatch: `calculate_paradox({"paradox": "transcendental_gamma_euler_transcendental"`

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